An HR organisation wishes to introduce a new set of training programs to increase the work efficiency of their employees. To do this, the organisation hires 3 vendors such that each vendor will train a group of employees with a one-week program. It is assumed that the employees across the three groups have same work efficiencies before the start of the program. The following table summarises the scores obtained post the one-week training. Each group has 8 employees.

Now, find out the mean efficiency of each vendor and determine which vendor shows a higher significance of improvement in employee efficiency statistically at the 95% level.

Group 1 scores (Vendor 1)	Group 2 scores (Vendor 2)	Group 3 scores (Vendor 3)
45	61	35
29	53	21
56	41	33
52	58	27
45	53	22
45	47	26
41	44	30
Adapted from: Rakshir, S. (2019). R for Beginners. India: McGraw Hill.		

The cf at 95% for vendor 1 is 44.71
The cf at 95% for vendor 2 is 51
The cf at 95% for vendor 3 is 27.71

This means that vendor 2 has the highest cf so this shows the higher significance of improvement at 51.

First I created the groups and assigned them names that was easy to identify.
I then ran mean tests, and then performed the t.test which calculated the confidence interval at 95%. The code for these tests are below.

```
> vendor1<-c(45,29,56,52,45,45,41)
> vendor2<-c(61,53,41,58,53,47,44)
> vendor3<-c(35,21,33,27,22,26,30)
> mean(vendor1)
[1] 44.71429
> mean(vendor2)
[1] 51
> mean(vendor3)
[1] 27.71429
```

```
> t.test(vendor1)

        One Sample t-test

data:  vendor1
t = 13.792, df = 6, p-value = 9.037e-06
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 36.78153 52.64704
sample estimates:
mean of x
 44.71429

> t.test(vendor2)

        One Sample t-test

data:  vendor2
t = 18.419, df = 6, p-value = 1.651e-06
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 44.22481 57.77519
sample estimates:
mean of x
       51

> t.test(vendor3)

        One Sample t-test

data:  vendor3
t = 13.881, df = 6, p-value = 8.707e-06
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 22.82879 32.59978
sample estimates:
mean of x
 27.71429

>
```